

MODULE 1

BIOMEDICAL ENGINEERING TECHNOLOGIST

Apply Human Anatomy and Physiology Principles

November/December 2025

SECTION A (40 MARKS)

Answer ALL questions in this section in the answer booklet provided.

1. The study of human body is divided into various subdivisions based on the structures under study. Identify the TWO main subdivisions of Anatomy. [2marks]
2. Medical terms are used by doctors and clinicians to communicate together in a medical field. Outline the meaning of the term arrhythmia as used in medicine. [2marks]
3. Ethics is a study of what is morally upright or wrong, just or unjust. Identify the FOUR fundamental Principles of Medical Ethics. [4marks]
4. The skeleton of an adult is made up of 206 bones. Define a skeleton and list FOUR functions of the skeletal system. [5marks]
5. Diabetes is a condition of human body associated with production of hormones. Highlight the difference between Diabetes mellitus and Diabetes insipidus. [4marks]
6. The five senses are the sensory mechanisms that allow humans and other animals to perceive and interact with their environment. Highlight these FIVE senses. [5marks]
7. Skin adaptation plays a crucial role in an individual's ability to tolerate extreme temperatures. The human skin is a dynamic organ that can adjust to various environmental conditions. Highlight FOUR examples of skin adaptations to extreme temperatures. [4Marks]
8. Human tooth is made up of various parts with different characteristics. Sketch and label the structure of the human tooth. [5Marks]
9. Urinary system is associated with various disorders. Highlight FIVE common disorders of the urinary system. [5Marks]
10. The skull is made up of flat bones joined together by joints.
 - i. Name THREE bones that make up the cranium also known as skull. [3Marks]
 - ii. Identify the type of joint that holds these bones together. [1Mark]

SECTION B (60 MARKS)

Answer any THREE questions from this section.

11. Recent advancements in cardiovascular medicine have significantly improved the diagnosis, treatment, and management of heart and vascular conditions.

- a) Explain FIVE of the latest breakthroughs that has helped in the management of heart and vascular conditions. [10Marks]
- b) Medical equipment used in the monitoring and treatment of the cardiovascular system plays a crucial role in diagnosing, managing, and treating heart and vascular conditions. Discuss FIVE common medical devices and equipment used for this purpose. [10Marks]
12. Tom had just finished servicing a refrigerator during a practical. While carrying out the performance test he accidentally touched the very hot compressor and quickly retracted his finger. Use this scenario to answer the following questions.
- a) Explain FOUR neural and muscular activities that occur in the body when one touches a hot object. [8Marks] b) Define the term reflex action. [2Marks]
- c) With aid of labelled diagram explain the structure of a neuron. [10Marks]
13. Knowledge of the respiratory system is crucial for an engineer working in the intensive care unit where machines like the ventilator and CPAP are used.
- a) With aid of a sketch, discuss the structure of the human respiratory system. [14Marks]
- b) Respiratory disorders are a group of medical conditions that affect the respiratory system, which includes the organs and structures responsible for breathing. Highlight SIX disorders associated with the respiratory system. [6 Marks]
14. The liver is the largest glandular organ of the body and considered an accessory structure to the entire digestive system,
- a) Highlight SIX major functions of the liver. [6 Marks]
- b) Track the path followed by food once ingested from the mouth until it's excreted from the body. [14 Marks]

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